

SEMINAR PRESENTATION

Extended Reality (XR) & Immersive Technologies

Bridging the physical and digital worlds through virtual, augmented, and mixed reality

CSA1017 – Software Engineering



Virtual · Augmented · Mixed Reality

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Introduction to the Emerging Technology

XR

An umbrella term uniting every technology
that blends the real and virtual worlds

1

Umbrella technology

XR covers Virtual, Augmented, and Mixed Reality under one collective term.

2

Merged environments

It fuses physical and digital environments into a single, unified experience.

3

Powered by innovation

Enabled by rapid advances in sensors, displays, and real-time computing.

4

Expanding fast

Moving quickly from niche gaming into mainstream enterprise adoption.

Why XR Is Needed

1

Demand for immersion

Users increasingly seek engaging, immersive digital experiences over flat screens.

2

Beyond 2D limits

Traditional screens fall short for complex training, design, and visualization tasks.

3

Remote collaboration

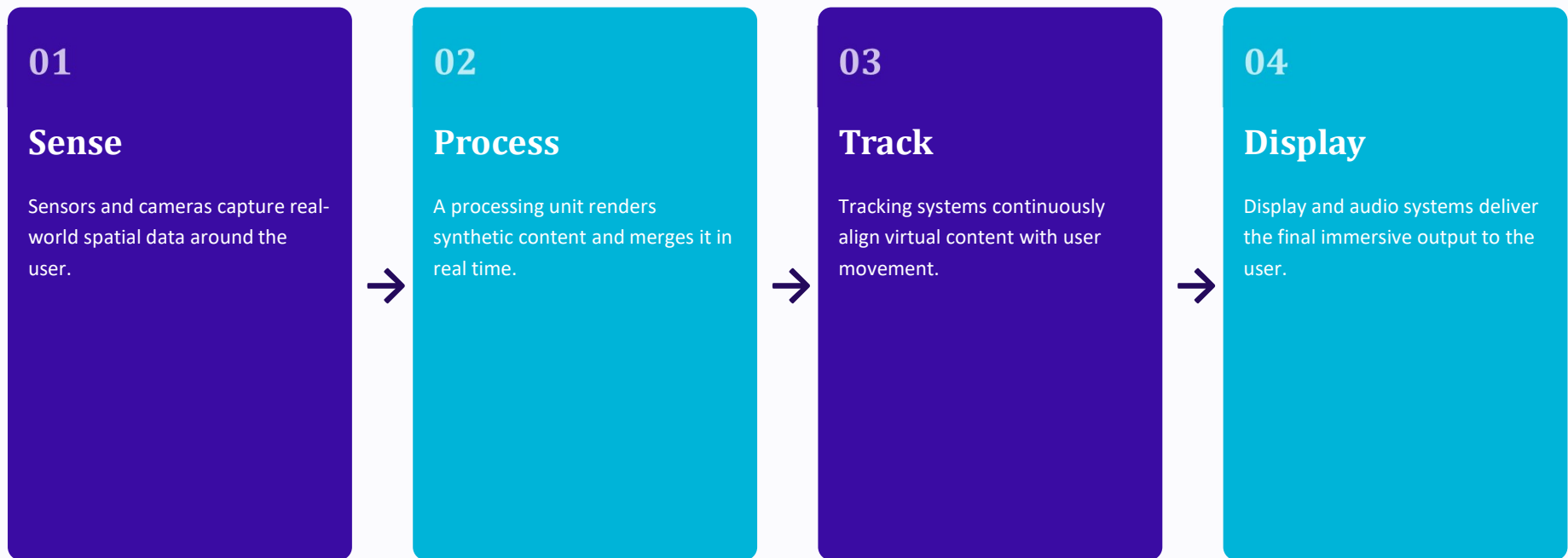
Distributed teams need richer tools to collaborate as if physically together.

4

Safer simulation

Industries want cost-effective, risk-free environments to simulate real scenarios.

Working Principle / Architecture



Key Components and Technologies

1

Headsets & smart glasses

Head-mounted displays (HMDs) and smart glasses deliver the visual experience.

2

Motion & haptic devices

Motion tracking sensors, controllers, and haptic devices add touch and movement.

3

SLAM algorithms

Simultaneous Localization and Mapping lets devices map spaces in real time.

4

GPUs & connectivity

High-speed GPUs paired with low-latency wireless links power smooth rendering.

Applications / Real-World Use Cases

1

Healthcare

Surgical training simulations and immersive pain-management therapy programs.

2

Education

Virtual laboratories and interactive historical or scientific field tours.

3

Manufacturing

Remote assembly guidance and digital twins for factory-floor operations.

4

Retail

Virtual try-ons and immersive product showcases for online shoppers.

Advantages and Limitations

ADVANTAGES

- ✓ Enhances learning retention and learner engagement significantly.
- ✓ Enables safe simulation of hazardous, high-risk scenarios.

LIMITATIONS

- X High hardware costs and reported motion-sickness concerns.
- X Ongoing privacy and data-security challenges for users.

Recent Developments and Trends

1

Lighter, sharper headsets

Standalone headsets are now lighter with noticeably improved display resolution.

2

AI-driven realism

AI integration produces more realistic avatars and dynamically generated environments.

3

Rise of the metaverse

Persistent, shared virtual spaces are emerging as the next social and work frontier.

4

Cloud-streamed XR

5G and 6G networks are enabling cloud-based streaming of high-fidelity XR content.

Challenges and Future Scope

CHALLENGES

1

Interoperability gaps

Lack of standardized content and protocols across XR platforms and devices.

2

High development costs

Steep production costs limit adoption among smaller organizations and creators.

FUTURE SCOPE

1

Brain-computer interfaces

Direct neural interaction could become a future input method for XR systems.

2

Everyday integration

XR is poised to blend seamlessly into everyday consumer devices and routines.

Case Study / Example

30%

reduction in training time reported using AR-guided assembly

1

Enterprise training with Meta Quest

Meta Quest headsets are deployed for large-scale enterprise training simulations.

2

Boeing's AR wiring guidance

Boeing uses AR overlays to guide technicians through aircraft wiring assembly.

3

Fewer errors, faster onboarding

Reported outcomes include reduced training time and fewer assembly errors.

4

Proven measurable ROI

The case demonstrates measurable return on investment across industrial settings.

Conclusion and References

1

Redefining interaction

XR is redefining how humans interact with digital information and each other.

2

Innovation drives adoption

Continued innovation in hardware and software will drive broader everyday adoption.

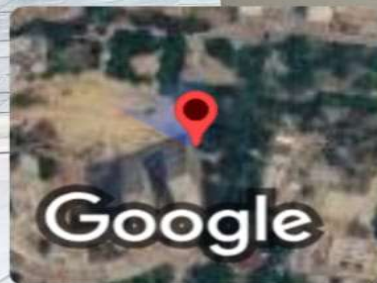
3

Affordability is key

Future success depends on affordability and a rich, accessible content ecosystem.

REFERENCES

IEEE VR & 3D User Interfaces journals · Gartner Emerging Technology Hype Cycle · IDC Worldwide AR/VR Spending Guide



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